

PLSC 504: “Topics in Political Methodology”

Exercise Three

September 28, 2026

Part I

Clive Granger won a Riksbank Prize in economics in part because of his work on time series. In 1974, building on work by French economist Edmond Malinvaud (1966), Granger and his coauthor Paul Newbold derived the idea of “[spurious regressions](#)”: the phenomenon whereby, because integrated series have a tendency to “wander,” it is often the case that a regression of one on the other will yield significant results (thus suggesting that the two series are related), even if the two series are in fact completely independent.

Your task in Part I is to discuss what happens when you fit an OLS regression two *unrelated* series. You will do this via simulation. More specifically:

1. Begin by generating two short (say, $T = 10$) independent I(1) series (say, Y and X):

$$\begin{aligned}X_t &= \rho X_t + u_t \\Y_t &= \rho Y_t + v_t\end{aligned}$$

with $\rho = 1$.

2. Regress Y on X :

$$Y_t = \beta_0 + \beta_1 X_t + u_t$$

3. Record (a) the P -value for the test that $\hat{\beta}_1 = 0$, (b) the R^2 statistic for the regression, and (c) the Durbin-Watson $\hat{d}$ statistic for first-order residual autocorrelation for that regression.
4. Repeat (1) - (3) many times, and display and discuss the distribution of these statistics and what they suggest about the regressions of Y on X .
5. Vary T (the length of the two series) systematically, and repeat steps (1) - (4) for varying values of T .¹ Discuss how the distributions of ts , P -values, R^2 s, and $\hat{d}$ s vary as the length of the series grows, and how this compares to what we would normally expect for two variables which are statistically independent of each other.
6. Finally: vary the persistence of the series. Repeat steps (1) - (5) for $\rho \in \{0.1, 0.5, 0.9, 0.99\}$. How does the false-rejection rate depend on ρ ? Is there a sharp break at $\rho = 1$, or does the problem appear gradually as ρ approaches one? What does this suggest about applied settings in which we cannot be sure whether a series has a unit root?

¹Consider starting with a value of T that would be considered very “short” for a time-series analysis (again, say, $T = 10$) and increasing it to a very long series (perhaps $T = 1000$). Note that it is usually a good idea to increase sample sizes in simulations geometrically rather than linearly. Also: R’s built-in `arma.sim` function is your friend here.

Part II

Beginning at least with Barnum (1985), researchers have endeavored to verify empirically Mr. Dooley's famous quote that “[th’ Supreme Court follows th’ iliction returns.](#)” The basic idea – going back at least to Dahl’s work in the 1950s, but arguably to the Federalist Papers as well – is that, because members of the Court are appointed by the president and confirmed by the Senate, their political preferences (and, consequently, their decisions) will never stay too far out of line with those of the elected branches (and, therefore, the public).²

Here I’ll ask you to add to those earlier efforts, using time series data. The main variable of interest is the percentage of decisions handed down in each October term of the U.S. Supreme Court which ruled in a conventionally left/liberal direction.³ The main (sole) predictor is Jim Stimson’s annual measure of [policy mood](#). In both instances, higher values indicate greater liberalism on the indicator. Both series are available annually, from 1952-2020; those data are available on the course [github repository](#) in the “Exercises” folder.

Your assignment is to assess whether and to what extent the two series (Supreme Court liberalism and public mood) are related, in both the short- and long-term. In doing so, use the approaches and methods we discussed in class.

This assignment is due *electronically*, as a *PDF file*, at 11:59 p.m. EST on Monday, October 12, 2026; you can submit your homework by emailing copies **both** to Dr. Zorn (zorn@psu.edu) and Ms. Bandeira (1fb5423@psu.edu). This assignment is worth 50 possible points.

²This being 2026, we have recently seen a few particularly egregious violations of this general principle. We cannot, apparently, have nice things.

³Details can be found at the Supreme Court Database [website](#).